

Enhancing the Accuracy and Explainability of Heart Disease Prediction Models through Interpretable Machine Learning Techniques

Owana Marzia Moushi

*Electrical and Computer Engineering
University of Nebraska - Lincoln
Omaha, Nebraska, USA
omoushi2@huskers.unl.edu*

Nishat Ara

*Pharmaceutical Sciences
Texas A&M University
College Station, Texas, USA
nishat.ara@tamu.edu*

Md Helaluddin

*Computer Science & Engineering
New Mexico Institute of Mining and Technology
Socorro, New Mexico, USA
md.helaluddin@student.nmt.edu*

Himadri Shekhar Mondal

*Member, IEEE
bdhimadri@ieee.org*

Abstract—Cardiovascular disease is a significant worldwide health concern that has a devastating impact on mortality rates around the globe. Leveraging machine learning techniques for cardiovascular diseases prediction has the potential to prevent further consequences significantly. This study proposed a Grid Search Cross Validation-based model to predict heart diseases. The dataset has been utilized in this study consists of 51.3% heart disease samples and the rest of them have healthy conditions. A combination of the Pearson Correlation Coefficient and Permutation Important algorithm is used for selecting the best features for the dataset. Several Machine learning models such as Multi-layer Perceptron, Logistic Regression, Extra Trees, Decision trees, Random forests, Native Bayes, k-nearest neighbors, and C5.0 have been applied in this research. This experimental analysis is focused on different machine-learning models with default parameters which give the best outcome of 94.63% accuracy by Random Forest model. The proposed Grid Search Cross Validation model with the selected best features has been used on top of the classic machine learning approaches, giving a significant accuracy gain of 98.57% by the Extra Trees classifier. The excellent proposed result suggests that applying the Grid Search Cross Validation on the classic machine learning approach might be a promising method for cardiovascular disease detection.

Index Terms—Cardiovascular diseases, Machine Learning, GridSearchCV, Feature Selection

I. INTRODUCTION

Cardiovascular diseases (CVD) are typically a consequence of aortic, coronary, peripheral arterial, and strokes and these lead to heart and blood-related disease. Nowadays, the increasing rate of heart disease is the main reason for high mortality rate in USA [1]. Due to an unhealthy lifestyle and demanding schedule with technology people often are responsible for developing heart disease [2], [3]. The World Health Organization (WHO) reports that each year, CVD is responsible for approximately 19.05 million deaths around the world [4]. Early prediction of heart disease can significantly increase the rate of survival.

Machine learning (ML) is a popular approach in this twenty century and is playing an important role in getting a more accurate prediction without explicit programming. It plays a vital role for the prediction of cardiovascular diseases with the help of well-trained dataset. Selecting features and removing over-fitting is an important criterion for getting better accuracy in machine learning models.

ML is utilizing for the prediction of heart related illness which is reported in several research. Currently, the majority of popular ML approaches such as Logistic Regression (LR), Random Forest (RF), Multi-layer Perceptron Classifier (MLP), Decision Tree Classifier (DT), XGBoost Classifier, AdaBoost Classifier, Stochastic Gradient Descent (SGD), Gaussian Classifier, k-nearest neighbors (KNN), Quadratic Discriminant Analysis, Gradient Boosting Classifier, Naive Bayes (NB), Support Vector Machine (SVM), C5.0, Ensemble Classifier models are utilized by researchers to predict heart disease [3], [5]–[8]. This paper highlights several crucial contributions, such as.

- A combination of two different feature selection algorithms has been used to determine the most important features for the selected dataset.
- Various classic ML models have been used to detect heart disease.
- Grid Search, Cross Validation (GridSearchCV) based ML models are being applied on top of classic ML models to improve performance.
- Performance has been analyzed for both cases by using different evaluation metrics.

This research is structured as, section II discusses the previous work in this field, section III explains the preliminaries that are necessary to conduct this research, section IV discusses the dataset for this work, section V explains the approaches that are used to predict the heart disease, section VI discusses the experimental analysis, and finally, section VII marks the end

of this work, and presents a summary of the main findings.

II. LITERATURE REVIEW

Significant research efforts are ongoing toward heart disease prediction that can reduce the potential number of death. In [9], authors proposed a CVD prediction system that involves data collection from various sources such as medical records and surveys using machine learning algorithms. It consists of data collection, feature selection, and prediction model components. The prediction model is trained using various machine learning algorithms such as LR, MLP, catboost algorithm, and RF algorithm, as well as a novel approach for the comparative analysis of the mentioned ML algorithms named hybrid Split Multi-Level Trunking (SMLT). The proposed system by the Hybrid SMLT algorithm achieved the highest impressive cumulative accuracy rate of 98.84%. However, the author did not show any precision and sensitivity while handling large datasets and analyzing interconnections. In [10], authors suggested a hybrid ML algorithm, the combination of NB and SVM algorithms to predict CVD using a dataset of critically ill heart disease patients. They utilized various ML algorithms including DT, LR, KNN, NB, and SVM algorithms and for performance - accuracy, precision, sensitivity, specificity, and classification error a range of metrics were evaluated. The accuracy rate of the proposed algorithm was 97.2%. In [11], authors described a bagging technology for heart disease prediction where the results of this innovative technique achieved the highest accuracy of 90.06% on the data sets compared with all individual classifiers. They used four different classifiers- KNN, DT, SVM, and RF for bagging to analyze how well their proposed model performed.

The authors of [3] used the prominent UCI heart disease dataset with the most relevant attributes and pre-processed the data for applying the SGD, KNN, RF, LR, and finally, Ensemble Classifier is used to combine all of them that gives the better accuracy. For all of the models, accuracy is improved when GridSearchCV is used to find the most important and optimized parameters except LR. In [5], authors tried 10 different types of ML models using voting or weighted approach that was utilized by [3] to predict the stroke patient from the collected data of a local hospital. The best accuracy was achieved from the weighted model. The authors of [2] used LR and KNN with an ensemble method which was previously experimented by the authors of [3], [5]. In [2], authors proposed an approach for data balancing with the utilization of the most important features to apply that ensemble method. For data balancing, they focused on the missing value replacement technique, handled the imbalance class, and removed outliers by using box plots. Three different datasets were used to measure the model's accuracy in all cases which is more than 95%. In addition, 3-fold, 5-fold, and 10-fold approaches were used for the validation.

III. BACKGROUND

Supervised-based classification models have been used for this work. A brief explanation of them is as follows: LR uses a

sigmoid function to obtain the predicted output from weights or coefficients of the input variable, NB assumes that every feature is independent of each other for predicting the output based on previous knowledge, KNN presents the data as a m dimensional vector where m is the number of features to calculate the distance between every point, at the next stage, the model labels the unvisited data based on the nearest visited data, DT splits the data set into a number of subsets by using the most important features and the classification is done on the tree leaves, RF is one type of ensemble method where trees are trained on the original data set by considering the random sample and also considering the major votes from the trees, SVM classifies the data sets based on the relationship border of positive and negative classes, C5.0 splits the sample by using maximum information gain, after the first sub-sample it splits the data set again by using a different area and continues the splitting until the model reaches the point where the dataset cannot be sampled again, and the low-level samples are removed as they have less contribution on the model, MLP classifier has one input, one output, and one or more hidden layers where many neurons can be stacked altogether, and the input features are mapped to the corresponding label of class, connection weights are adjusted by using an optimization algorithm during training, and Extra Tree (ET) Classifier trains multiple decision trees and combined the results from those trees for making predictions.

In addition, GridSearchCV was used for determining the best hyperparameters of the classic ML models. Basically, this GridSearchCV tries all possible values from the dictionary and checks the performance of the model by applying the Cross-Validation (CV) method. For every hyperparameter set, the model can calculate the accuracy which actually chooses the best parameter [12].

IV. DATASET

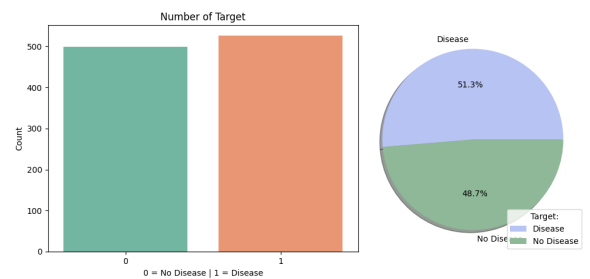


Fig. 1: Dataset Visualisation

The chosen dataset for this study comprises four distinct databases namely Long Beach V, Switzerland, Hungary, and Cleveland which initially had 76 attributes. Among all the attributes, only a subset of 14 attributes such as serum cholesterol, sex, resting blood pressure, and fasting blood sugar, chest pain type, provided accurate predictions during the examination. The dataset's target variable refers to the presence of CVD in the subject where the disease and without disease conditions are indicated by 1 and 0, respectively. In

the dataset, about 51.3% of the samples are disease class and the rest 48.7% is non-disease class as shown in Fig. 1 [13].

A. Removing redundant records

Redundant data has been removed from the data frame where duplicate rows were identified by comparing a specific row with all the columns in the dataset by using pandas. Afterward, the algorithm dropped the redundant rows by sorting the data frame rows on a single column in descending/ascending order. Finally, the duplicated data were removed by sorting the same data frame from multiple columns.

B. Feature Selection

Feature selection in the ML process is used to build optimized models for a specific dataset. A combination of Pearson's correlation coefficient matrix and permutation importance is utilized in this work to select the best sets of features. Pearson's correlation coefficient is commonly utilized for determining the correlation between features in the linear dataset. The algorithm measures the association between features (negative, positive, and neutral) ranging from -1 to 1, where a score of -1, 0, and +1 indicate fully negative linear, no, and positive correlation, respectively [14].

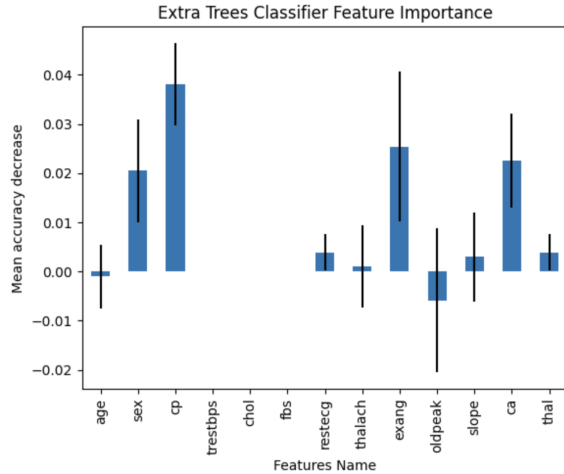


Fig. 2: Important Features for Extra Trees Classifier by Permutation importance algorithm

Permutation importance [15] is also utilized in this work to identify the important features of a specific type of ML model as an example shown in Fig. 2. This model-independent method gives a straightforward and interpretable way to look at the relative importance of individual feature in the model. This is a computationally expensive method due to it calculates for each feature how heavily the model relies on it and ranks it based on that. A combination of the two feature selection algorithms is employed to select the top 10 most relevant features for the proposed study

In addition, feature scaling, filling the missing values, and outlier detection task was done on the dataset before splitting it into the train-test set for ML models.

V. APPROACHES

GridSearchCV approach is utilized to enhance the model's performance and identify the optimal hyperparameters for each nine different ML models. Fig. 3 shows the proposed workflow, and the ML pseudocode is presented in Algorithm 1. CV is a statistical technique for measuring and improving the

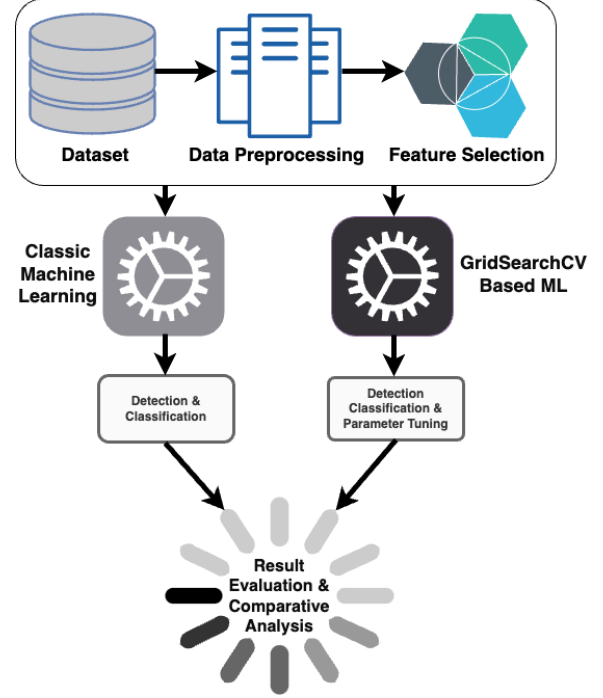


Fig. 3: Proposed Cardiovascular Diseases Prediction System

efficiency of an ML model by dividing the given dataset into k non-overlapping folds (k -fold) where k 's value is between 5-10; in this proposed work $k = 5$. Next, train the model with $k - 1$ folds and evaluate it with the other folds. The above approach is reiterated k times until each fold is considered as a test dataset and remains as training. The CV means accuracy is calculated by averaging out the losses across all k folds. This is a continuous process (repeats k times) until all folds are considered as a validation set at least once. Finally, the performance metric is evaluated by averaging the losses across all k folds as shown in Eq. 1.

$$CV = \frac{1}{K} \sum_{i=1}^K L_i \quad (1)$$

where, CV = cross-validation, k = number of folds, $\sum L_i$ = sum of the losses across all folds

GridSearchCV performs an exhaustive search over a predefined hyperparameter space to find the optimal hyperparameters for a given model. The algorithm for GridSearchCV is as follows-

- 1) Define the model name and the range of the hyperparameters grid.
- 2) Split Train and Validation Dataset.

TABLE I: Best Chosen parameters through GridSearchCV

Predictive Model	Parameters	Hyperparameters
LogisticRegression	C = 10	C = 10 penalty = l2
DecisionTreeClassifier	max_depth = 9 min_samples_split = 10	max_depth = 9 min_samples_split = 10
SVM	C = 10 $\gamma = 0.1$ probability = True	C = 10 $\gamma = 0.1$ kernel = rbf
RandomForestClassifier	n_estimators = 200	n_estimators = 200
KNeighborsClassifier	metric = manhattan n_neighbors = 10 weights = distance	metric = manhattan n_neighbors = 10 weights = distance
NaiveBayes	var_smoothing = 3.5112e-05	var_smoothing = 3.5112e-05
MLPClassifier	activation = tanh hidden_layer_sizes = 64	activation = tanh hidden_layer_sizes = 64 learning_rate = constant solver = adam
ExtraTreesClassifier	criterion = entropy max_depth = 20	criterion = entropy max_depth = 20 min_samples_leaf = 1 min_samples_split = 2 n_estimators = 100
C5.0	max_depth = 10	max_depth = 10 min_samples_leaf = 1 min_samples_split = 2

- 3) For each combination of hyperparameters train the model and evaluate.
- 4) Select a set of hyperparameters that gives the best performance on the validation set.
- 5) Select chosen hyperparameters for train the model.
- 6) Measure the performance.

Algorithm 1: Algorithm for Heart Disease Patient Prediction

Input: Heart Disease Dataset

Output: Produce the final models, chosen hyperparameters, and test performance of heart disease prediction

Define the ML_models and their hyperparameter_grid
Split dataset into training and validation sets (X_{train} , X_{val} , Y_{train} , Y_{val})

$L \leftarrow []$ #L is an empty list

```

for model_name in ML_models: do
    best_performance  $\leftarrow -\infty$ 
    best_hyperparameters  $\leftarrow None$ 
    for hyperparameters in hyperparameter_grid: do
        model = CreateModel(model_name,
                             hyperparameters)
        model.fit( $X_{train}$ ,  $Y_{train}$ )
        val_performance  $\leftarrow$  evaluate_model(model)
        if val_performance > best_performance:
            best_performance  $\leftarrow$  val_performance
            best_hyperparameters  $\leftarrow$  hyperparameters
    final_model = CreateModel(model_name,
                              best_hyperparameters)
    final_model.fit( $X_{train}$ ,  $Y_{train}$ )
    test_performance  $\leftarrow$  EvaluateModel(final_model)
    L.input  $\leftarrow$  {final_model, best_hyperparameters,
                  test_performance}

```

Return L

The purposes of augmenting GridSearchCV and CV are

to enrich the model's performance and find the optimal hyperparameters which also significantly improve a given model's performance on the test set. GridSearchCV may incur significant computational costs, particularly, when it deals with large datasets or complex models but for small datasets, it is a robust method for finding the optimal hyperparameters for a given particular model.

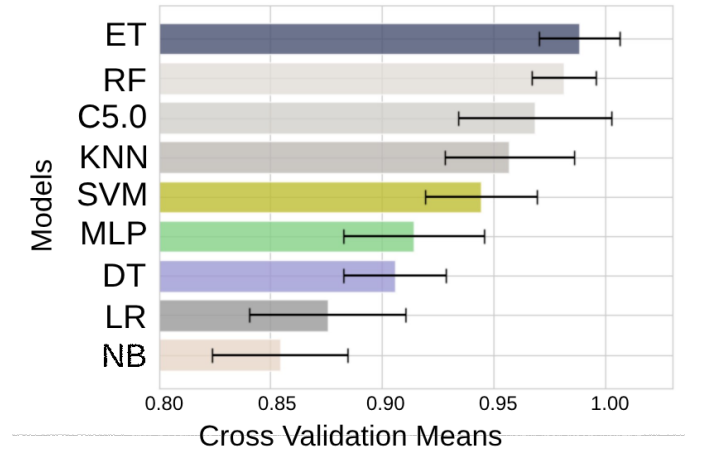


Fig. 4: Cross Validation Means with Cross Val Errors i.e., standard deviation of the performance metric across 5-folds.

VI. EXPERIMENTAL RESULT ANALYSIS

This section will explain the results of classic and proposed GridSearchCV-based ML models which helps to gain better accuracy over conventional models. Generally, the performance of an ML model can be captured in different ways and several factors can be taken into account. This work utilized different evaluation metrics for prediction tasks namely accuracy, precision, sensitivity/recall, specificity, F1-score as shown in Eq. 2, 3, 4, 5, 6 and area under ROC curve. Those metrics will help to understand the model's ability and how it performs on a specific dataset. The different evaluation metrics for the proposed work are shown in Fig. 5.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

$$Sensitivity/Recall = \frac{TP}{TP + FN} \quad (4)$$

$$Specificity = \frac{TN}{TN + FP} \quad (5)$$

$$F1Score = \frac{2 * Recall * Precision}{Recall + Precision} \quad (6)$$

where TP = True Positive; TN = True Negative; FP = False Positive; FN = False Negative, Sensitivity/Recall represents the correct prediction rate, Precision represents the actual prediction rate, Specificity represents the True Negative Rate (TNR) and F1-Score represents the weighted mean(average) of the precision and recall.

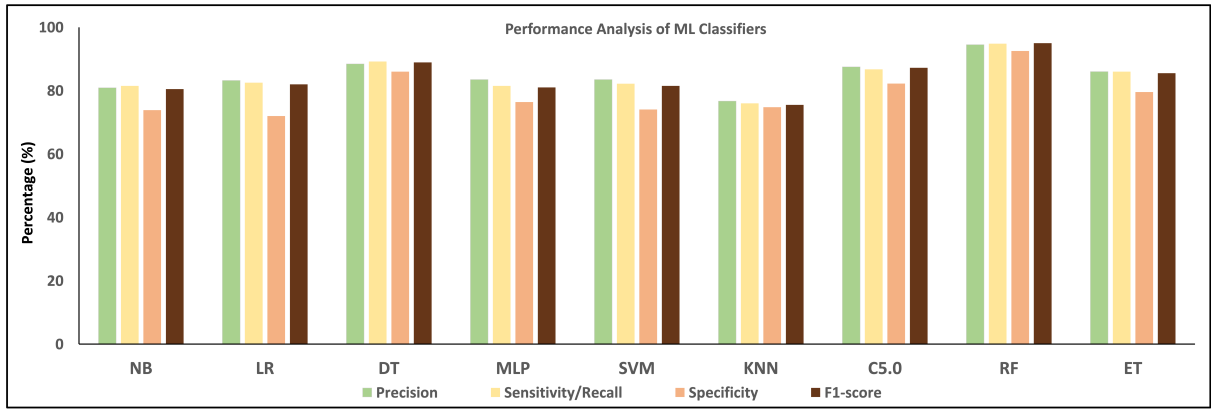


Fig. 5: Different performance metrics of various classic ML algorithms

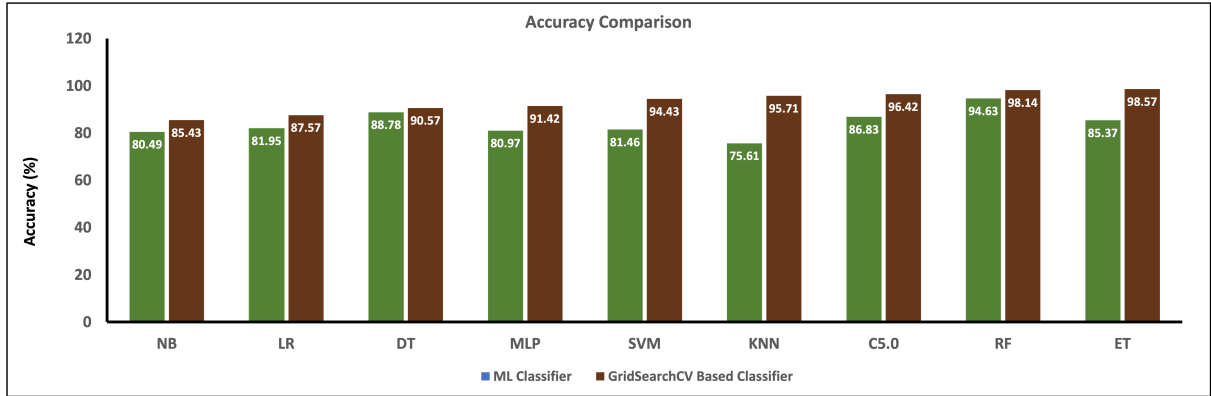


Fig. 6: Accuracy Comparison of Various Classic ML Algorithms with Proposed GridSearchCV-based Method

Our results demonstrated that GridSearchCV with fine-tuning parameters shown in Table I outperformed the classic ML model's performance from scratch, highlighting the effectiveness of parameter selection for this task. The mean cross-validation accuracy across all models with a standard deviation over 5-fold is shown in Fig. 4.

The most effective model was ExtraTreesClassifier, which achieved an accuracy of 98.57%, followed by RF at 98.14%, and C5.0 at 96.42% as shown in Fig. 6 and table I. The other models, in descending order of performance, were KNN, SVM, MLP, DT, LR, and NB.

In contrast, when not applying GridSearchCV, the top-performing model was RF with a classification accuracy of 94.63%, followed by DT at 88.78%, and C5.0 at 86.82%. This highlights the importance of utilizing fine-tuned parameters to improve the performance of ML models.

The excellent accuracy data for detecting cardiovascular disease by our proposed algorithm GridSearchCV indicates it, as a promising reliable algorithm. Our obtained detection accuracy percentage (98.57%) correlates with the previous findings 98.84% by Hybrid SMLT [9]. On the other hand, our result outperforms existing peer work that obtained 90.06% by bagging ensemble technique [11], 97.2% by Hybrid NB+SVM [10], 83% by Random Forest Algorithm [16], 87% by

KNN [17], and 92.37% by FCMIM-SVM [18].

Overall, our findings demonstrate the importance of choosing appropriate parameters for a model and fine-tuning them using GridSearchCV for improved performance. After a rigorous testing and training process, the outcome implies such the GridSearchCV has better accuracy when compared with other algorithms evaluated.

VII. CONCLUSION

The unhealthy lifestyle of modern people plays a big role in the rise of heart disease cases. Early Prediction of heart disease can save a lot of people. Several ML-based models were employed in this study to forecast people with heart disease. Additionally, GridSearchCV has been applied with the classic ML model to significantly improve the model's accuracy by tuning the parameters. Pearson's correlation coefficient and Permutation importance were used in combination to find out the best features which also contribute to improving the accuracy. RF presented the best accuracy over all the classic ML models while ET gave the superior performance for the GridSearchCV ML models. The next phase of the work is to compare the proposed GridSearchCV ML model on different datasets as well as will try to improve the presented work.

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